06.10.2025

Task 8: Normalizing databases using functional dependencies upto BCNF

AIM:

To normalize the Hotel management system database upto Boyce-codd Normal Form (BCNF) using functional dependencies, reducing redundancy and ensuring data consistency

Step 1: Identify Entities and Attributes

Hotel : Hotel_ID, Hotel_Name, Location, Rating

Room: Room_ID, Room_Type, Rate, Hotel_ID

Customer: Customer_ID, Customer_Name, Phone, Email, Address

Booking: Booking_ID, Customer_ID, Room_ID, CheckIN_Date, Check Out_Date, Payment_Status)

Step 2: Determine Functional Dependencies

1. Hotel_ID → Hotel_Name, Location, Rating
2. Room_ID → Room_Type, Rate, Hotel_ID
3. Customer_ID → Customer_Name, Phone, Email, Address
4. Booking_ID → (Customer_ID, Room_ID, CheckIn_Date, Check_Out_Date, Payment_Status)
5. Room_Type → Rate (if same type rooms have identical rates)

Unnormalized Table

Booking_ID	Customer_Name	Phone	Location	Room_Type
B01	John	9876543210	Kochi	Deluxe, Standard
B02	Maria	8765432109	Sea view	Chennai

Repeating values in Room_Type violate 1NF

First Normal Form (1NF)

Remove repeating and multi valued attributes. each field must contain only atomic values.

Booking_ID	Customer_Name	Phone	Location	Room_Type
B01	John	9876543210	Kochi	Deluxe
B01	John	9876543210	Kochi	Standard
B02	Maria	8765432109	Chennai	Suite

All columns hold single atomic values

Table is in 1NF

Second Normal Form (2NF)

1. Table must be in 1NF
2. Remove partial dependencies

Decomposition to 2NF

Hotel Table

Hotel_ID	Hotel_Name	Location	Rating
H01	Blue Heaven	Kochi	4.5
H02	Sea view	Chennai	4.2

Customer Table

customer_ID	customer_name	Phone	Email	Address
C01	John	9876543210	john@gmail.com	Kochi
C02	maria	8765432109	maria@gmail.com	chennai

Room Table

Room_ID	Room_Type	Rate	Hotel_ID
R01	Deluxe	2000	H01
R02	Standard	1500	H01
R03	Suit	3000	H02

4. Booking Table

Booking_ID	Customer_ID	Room_ID	Check_in_date	Check out date	Payment
B01	C01	R01	10-10-2025	12-10-2025	Paid
B01	C01	R02	10-10-2025	12-10-2025	Paid
B02	C02	R03	11-10-2025	13-10-2025	Pending

Result: No partial dependencies → Database is 2NF

Third Normal Form

Must be in 2NF

Remove transitive dependencies

Decomposition into 3NF:

Hotel (Hotel_ID, Hotel_Name, Location, Rating)

Room (Room_ID, Room_Type, Hotel_ID)

RoomType (Room_Type, Rate)

Customer (customer_ID, customer_Name, Phone, Email, Address)

Booking (Booking_ID, customer_ID, Room_ID, CheckIn_Date, CheckOut_Date, Payment_Status)

All transitive dependencies removed → Database is in 3NF

BCNF

For every FD(X→Y), X must be a candidate key

Room_Type → Rate has no determinant it violates BCNF

To Fix BCNF violation

Decompose Room Table:

Room (Room_ID, Room_Type, Hotel_ID)

RoomType (Room_Type, Rate)

Now every determinant is a candidate key

VEL TECH	
EX No.	8
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	-
TOTAL (20)	15
SIGN WITH DATE	

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RESULT:

~~Thus the~~ normalized database using functional dependancy upto BCNF executed successfully